

# Functional Requirements

$D_{K_s}(E_{K_p}(m)) = m$  and  $D_{K_p}(E_{K_s}(m)) = m$  for every pair  $(K_p, K_s)$

- ✓ Generating a pair  $(K_p, K_s)$  is easy to compute (polynomial)
- ✓ Encryption is easy to compute (either polynomial or linear)
- ✓ Decryption is easy to compute (either polynomial or linear)
- Finding a matching key  $K_s$  for a given  $K_p$  is hard (exponential)
- Decryption without knowing the corresponding key is hard (exponential)

# RSA - Rivest, Shamir and Alderman

|                         |                                                                                                                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Key Size                | 1024 - 4096                                                                                                                                                                                       |
| Speed                   | <p>~ factor of <math>10^6</math> cycles / byte</p> <ul style="list-style-type: none"><li>• Key generation: 10 - 100 ms</li><li>• Encryption: 0.2 - 2 ms</li><li>• Decryption: 5 - 10 ms</li></ul> |
| Mathematical Foundation | Prime number theory                                                                                                                                                                               |